A FIFTEEN-MINUTE INTRODUCTION

Aethic Reasoning

Determinacy as a relation to an information state, and what the quantum measurement problem looks like once it is.

Ajax Benander · aethus.org

Framework: *Aethic Reasoning*, with the extraction papers on determinacy, counterfactuals, causality and decision

THE IDEA · IN ONE LINE

Instead of making quantum mechanics fit reality, make reality more like quantum mechanics.

THE USUAL DIRECTION

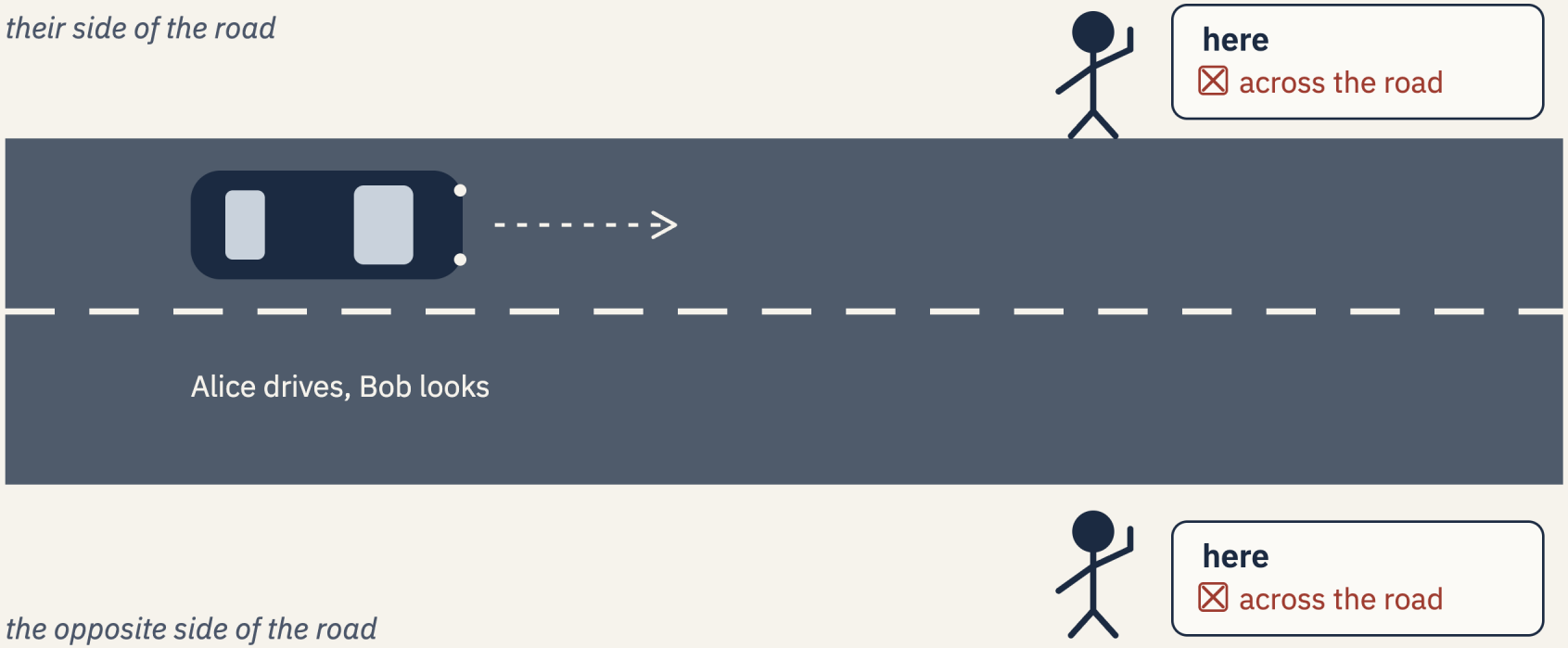
Fit the theory to a world of one-place facts, and pay in inventory: a collapse law, worlds, hidden variables.

THIS TALK

Give reality the structure the theory already has: alternatives co-present, determinacy relative to a record. See what it buys, and itemize the bill.

THE IDEA · ON A BACKROAD

Two hitchhikers, one road, no branch



BOB BEFORE

which side?

blank, not unknown

looks out the window

BOB AFTER

their side

stated

Relative to Bob before, the hitchhiker stands on both sides of the road, even now. So why do the two copies of Alice and Bob never look across the road and see each other?

THE IDEA · THREE PRINCIPLES, READ OFF THE ROAD

The other side is a ledger entry, not a branch

EXTRUSION

Relative to Bob after, the hitchhiker is on their side. Relative to Bob before, the hitchhiker is in superposition of both sides of the road, even now.

CENTRIC UNFOLDING

At the moment of looking, Bob was equally likely to land in the other after Aethus, hitchhiker across the road, though the hitchhiker stood there before the car arrived.

THE UNION PRINCIPLE

Each after Aethus declares the other position false inside itself, a ledger entry rather than an exterior branch: a **disagreeing** superposition. Leave the negations unstated: an **agreeing** superposition, proposed to yield quantum mechanics.

Bob before holds both, coupled: **Hitchhiker right × Alice right + Hitchhiker left × Alice left**

Alice's record is a record of the road, so the mismatched products are jointly invalid; when Bob looks, the remaining other term is struck.

THE IDEA · A COUNTERFACTUAL IS A CO-PRESENT ALTERNATIVE

What could have been is still there, relative to before

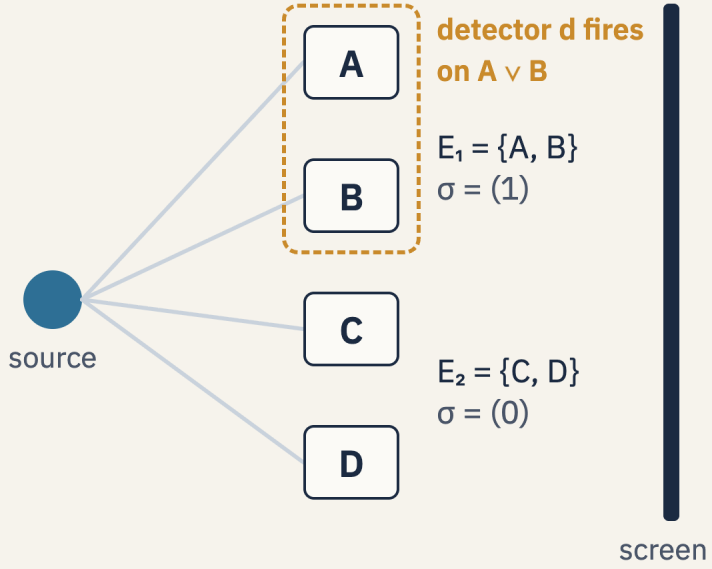


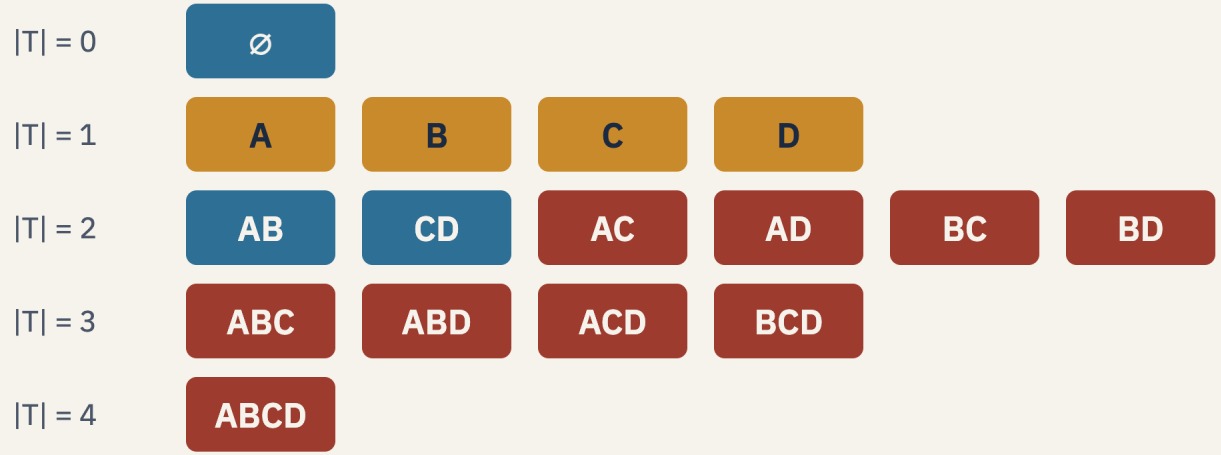
A counterfactual is a co-present alternative seen from **after** a record. A superposition is the same alternative seen from **before** one.

Now the physics, with the negations left out: four slits are four pockets with no entries between them. **Which sets of them stand together as one superposition?**

THE PAYOFF · THE POWERSET INVALIDATION THEOREM

The quantum–classical boundary is a partition, not a scale





All 16 claims "the traversal is coherent over exactly T", sorted by two rules and nothing else: no size, no mass, no observer.

✓ valid: T = ∅ or T = one whole class
no refinement can expose a contradiction

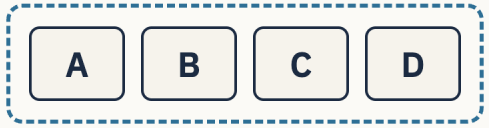
× rule 2: a proper part of a class
a line drawn where no record ever could; over-specification

× rule 3: spans two classes
some record separates them in every future; dead now, not later

THE PAYOFF · ERASER AND DELAYED CHOICE

Move a record, move the boundary

NO DETECTOR



one class

valid: **ABCD** — fringes across all four

"But I can aim the beam at A." Yes: the collimator is a record, and {A} becomes a class of the new partition.

DETECTOR ON A ∨ B



two classes

valid: **AB** **CD**

— fringes within each pair only

The record need not have been read. Its obtainability in every refinement kills ABCD now.

RECORD ERASED



one class again: no refinement can ever state the which-way record

valid: **ABCD** — the eraser

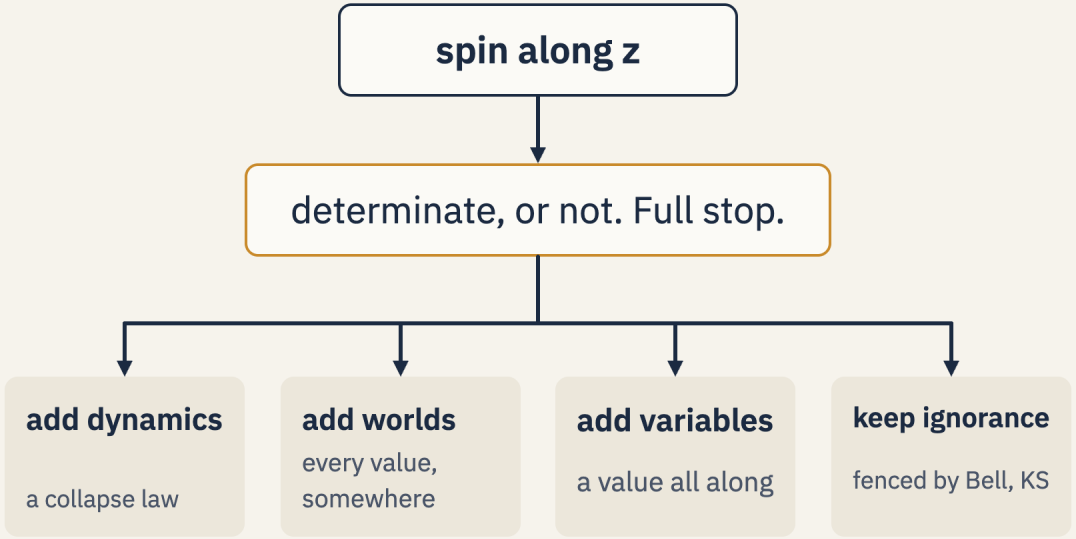
Decided after traversal? Irrelevant: the classification is tense-free. A record that will never be readable was never a record.

Double slit, quantum eraser, delayed choice: **one theorem, re-run on three partitions.** So where do the two rules come from?

WHERE THE RULES COME FROM · DETERMINACY IS TWO-PLACE

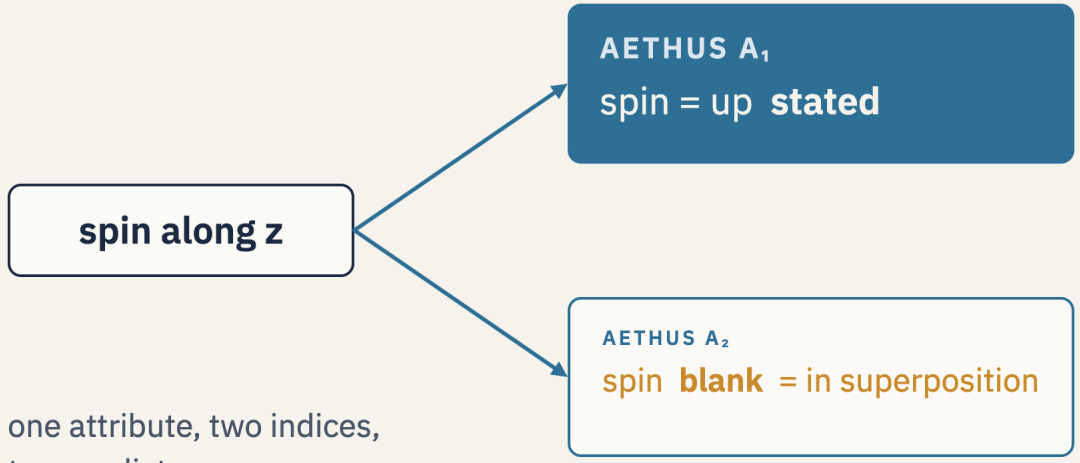
The measurement problem is a category mistake

STANDARD READING: DETERMINACY IS ONE-PLACE



Every answer keeps the notion and adds inventory.

AETHIC READING: DETERMINACY IS TWO-PLACE



one attribute, two indices,
two verdicts

Change the notion instead: a value is **relative to an information state**.

WHERE THE RULES COME FROM · THE OBJECT

An Aethus: a closed partial ledger, with a tree beneath it

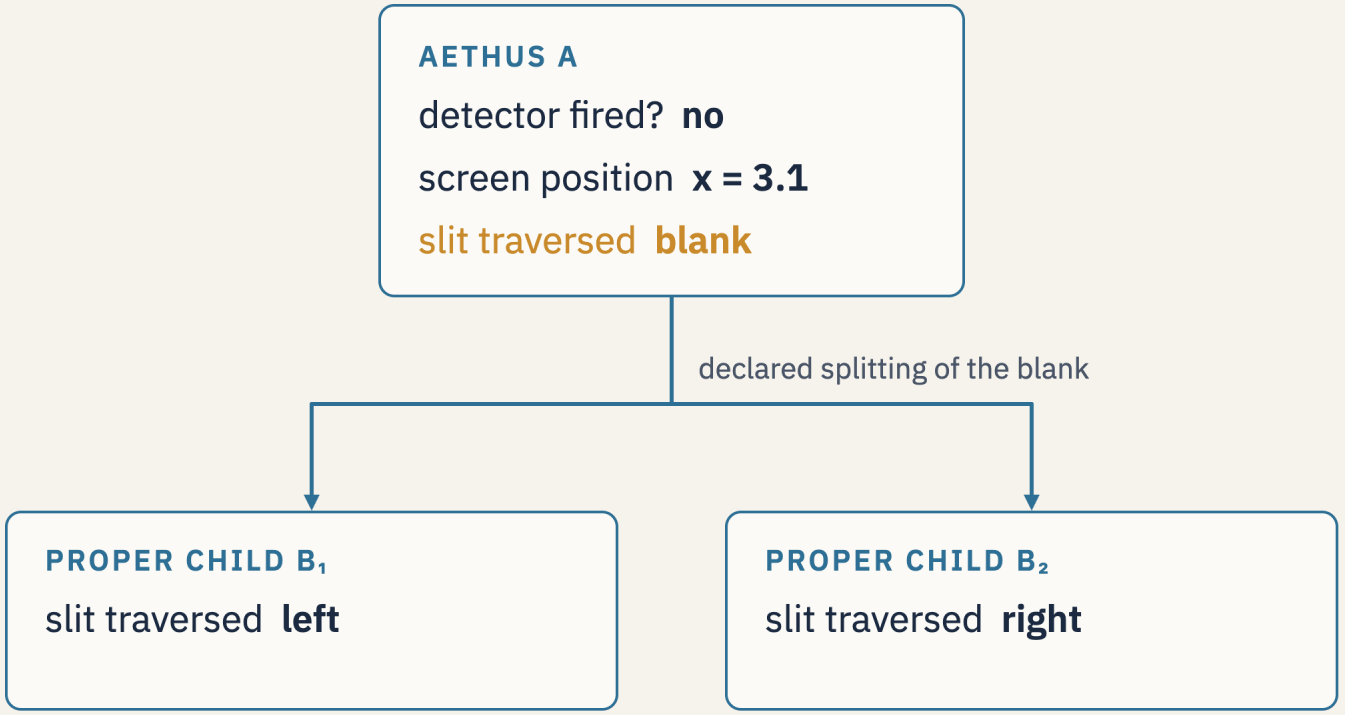
stated
a record fixes the value

blank
no record; open, not unknown

closed
carries its own consequences under the standing laws

proper child
a blank resolved along a declared split

invalid (base)
stated content meets a declared joint invalidity



B ≤ A, the Aethic tree: the order the third postulate quantifies over

WHERE THE RULES COME FROM · THE THREE POSTULATES

Three postulates, stated exactly

FIRST

Any attribution of reality is a statement of relation to a particular Aethus.

SECOND

If an Aethus is valid, then any given attribute will be in Aethic superposition relative to it if and only if it is nonpresent to it.

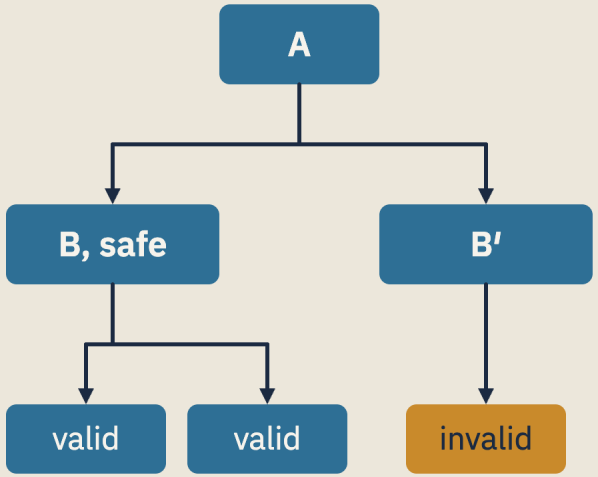
THIRD

If some Aethus is valid, then there exists a proper child Aethus to it for which every one of its own proper child Aethae are valid.

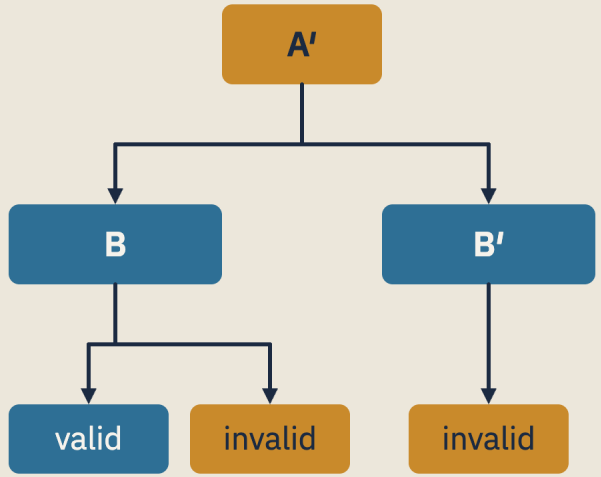
$$V[A] \Rightarrow \exists B \leq A \ \forall C \leq B : V[C]$$

An implication, not a biconditional: V ⊆ vF always; V = vF only on base generation.

∃ A SAFE CHILD, ∀ ITS CHILDREN VALID



A valid. One safe child is enough.



A′ invalid. No safe child: checkmate.

WHERE THE RULES COME FROM · THE UNION PRINCIPLE, STATED GENERALLY

Jointness is primitive; exclusion is an achievement

AGREEING SUPERPOSITION

one jointness: left ∨ right

left
interior states nothing about right

U

right
interior states nothing about left

cross content undeclared → the alternatives are co-performed → **interference**

DISAGREEING SUPERPOSITION

the same jointness, exclusions written in

left, ¬right
a closed room

×

right, ¬left
a closed room

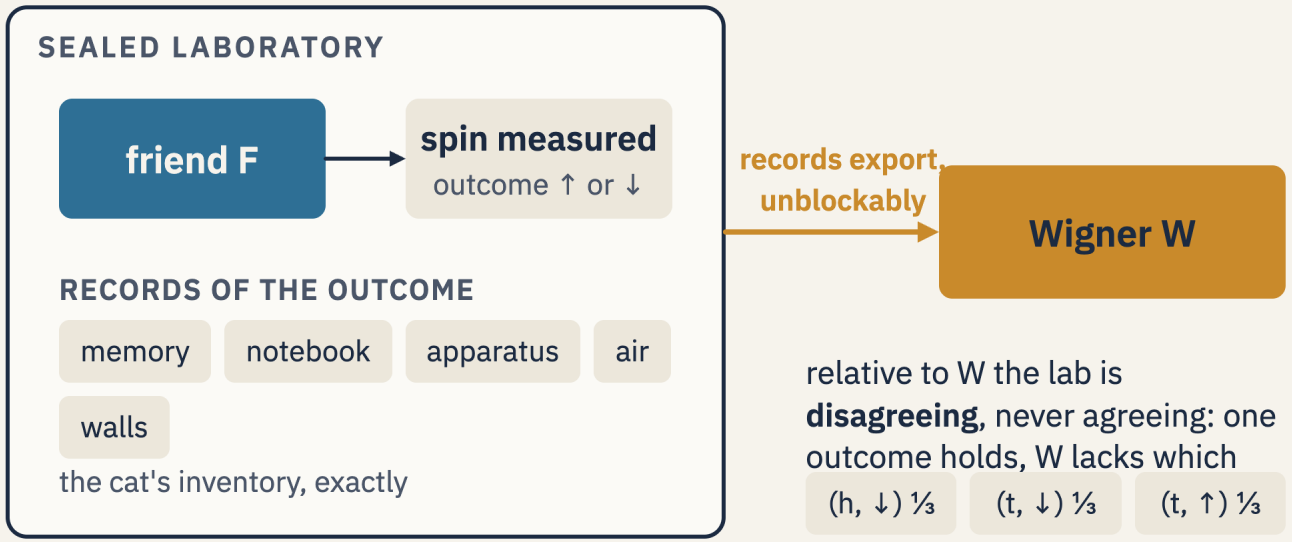
exclusions stated → weights over closed rooms → **the classical "or"**

The difference is written **in the alternatives**, not in the logic that composes them. A blank attribute is an agreeing superposition over its answer space.

Agreeing / disagreeing is qualitative and prior to any weight or phase; coherent / decoherent is quantitative, a property of a realization. No biconditional between them.

WHAT ELSE THE POSTULATES DECIDE · WIGNER'S FRIEND

Wigner's friend: records forbid the cut, not scale



FRAUCHIGER–RENNER, JOINT OUTCOMES

	coherent lab	Aethic mixture
(ok_1, ok_2)	**1/12**	**1/4**
(ok_1, $fail_2$)	1/12	1/4
($fail_1$, ok_2)	1/12	1/4
($fail_1$, $fail_2$)	3/4	1/4

The certainty step ($ok_1 \Rightarrow$ spin up) needs the coherent assignment; the mixture never starts the chain. The (ok_1, ok_2) cell is the experiment.

The one divergence from universal unitarity: **an observer-grade system exports its records unblockably.** Press here.

WHAT ELSE THE POSTULATES DECIDE · THE DEFLATION REDUCTIO

A blank is not an unknown

"BLANK" READ AS "UNKNOWN VALUE" MEANS

D1 noncontextual hidden values
one value per attribute, whatever else is measured

D2 verdict preservation
no classical verdict changes

D3 statistics = hidden-value frequencies
one measure μ; measurement independence

→

FINE'S THEOREM

a joint distribution with those marginals exists ⇔ CHSH holds

$|S| \le 2$

→

THE SINGLET, FROM THE KERNEL

$|S| = 2\sqrt{2}$

No lossless deflation exists.
The blank was constructive after all.

(a) Bell-violating sector
the agreeing regime; no single sample space

(b) Bell-bounded sector
the deflation exists: the classical filtration

The dilemma
deflate everywhere and fail, or halt exactly where the statistics exceed the bounds

WHAT ELSE THE POSTULATES DECIDE · COLLAPSE

Collapse is a transit between indices, not an event

AGREEING

superposition

cross content undeclared; interference

measurement →

a separating record becomes obtainable in every refinement (P3)

DISAGREEING

superposition

closed pockets, exclusions written in; Aethically fixed

observation →

an index acquires the record: selection of a pocket, a transit between Aethae

STATED

one value

relative to the updated Aethus

Only disagreeing superpositions ever collapse.

Nothing coherent collapses; observers are never in agreeing superposition.

Measurement ≠ observation.

The Sifting Principle: one is governed by accessibility, the other by acquisition.

Tense-free.

A retrieval later reads as a retrieval now; the past is as open as the future.

THE SETUP, RETURNED · COUNTERFACTUALS AS A SEMANTICS

The same engine runs counterfactuals

"IF X WERE THE CASE…"

1 **reopen**
the target to a blank intermediate, supports cleared

↓

2 **restate**
the antecedent's value

↓

3 **re-prune**
surviving pockets = the would-consequences

AETHUS A | s stated | law: s → y | {x, s, ¬z} invalid | x, z blank

perturb by x

s rests on nothing cleared → kept → **(x, s, y, z)** → **X ⊢~ y and X ⊢~ z**

perturb by x ∧ y

y's support s is cleared → not restored → **(x, ¬s, y, ¬z)** **(x, ¬s, y, z)** **(x, s, y, z)**

¬z survives: X ∧ y ⊬~ z

Cautious Monotony fails → not preferential → not Lewis, not Stalnaker, no smooth similarity ordering.
Machine-checked on the witness. The intermediate is the second postulate's blank; the pockets are what closure leaves.

THE BILL · EVERYTHING LOAD-BEARING

Everything load-bearing, on one ledger

POSIT	GRADE	BUYS
1 the index	posited	two-place determinacy; the paradox set dissolved
2 the constructive blank	posited	the open past; the whole counterfactual semantics
3 primitive jointness of "or"	posited, from the cases	agreeing / disagreeing; the classical limit
4 ∃∀ validity	derived in form, posited in class	checkmate; the Powerset Invalidation Theorem
5 one composition primitive	posited	the tree and its reduced form
6 zerosumfree semiring	design	validity separated from weight; nothing cancels a valid alternative
7 the kernel eiS/ħ	the one quantitative posit	Dirac–von Neumann recovered on the finite path-kernel domain
8 the realization bridge	a principle	"invalid" ⇒ "does not occur"

Not on the ledger, because derived: the projection postulate, no-signaling, the Born rule.

THE BILL · WHAT WOULD BREAK IT

What would break it

A superposition that survives a record every refinement can read

refutes the third postulate

A cross-term surviving a record, or vanishing without one

refutes the Union Principle

The Bell statistics from a single sample space

refutes the constructive blank

An observer-grade laboratory coherently reversed: (ok_1, ok_2) at 1/12, not 1/4

refutes the record premise, the one divergence from universal unitarity

Not a new mechanism. A changed notion of determinacy, with its bill itemized.